



[Outline](#) > 2. Improving the model > 2.3 Reaction time (Euler's Method) > Validation

Validation

Bookmark this page

In the previous section you have found an answer to the problem

Find t_{eq} such that $P(t_{\text{eq}}) = 720$, where

$$\begin{cases} \frac{dP}{dt} = 0.7 P \left(1 - \frac{P}{750} \right) - 20, \\ P(0) = 30, \end{cases}$$

which turned out to be $t_{\text{eq}} = 24.63$. You did this by using Euler's Method with $\Delta t = \frac{1}{32}$.

We can now state that for our problem, the rainbowfish population is close to the equilibrium size within **25** days.

Exercise A

1/1 point (graded)

We have chosen the stepsize of $\Delta t = \frac{1}{32}$ because then the value at the steepest part of the function (at $t = 13$) is correct within **20** fish. There we were looking at differences in P : vertical differences between the different curves in the graph.

To find the (rounded) value of $t_{\text{eq}} = 25$, we need an accuracy in time: horizontal differences between the curves in the graph. Calculate t_{eq} again, but now using the simulations with stepsizes $\Delta t = \frac{1}{4}$, $\Delta t = \frac{1}{2}$ and $\Delta t = 1$.

Which of the statements is true?

To calculate that the rounded value of t_{eq} equals **25**,

Choose all statements that are true.

☒ simulations with $\Delta t = 1$ give the same answer. ✓

☐ simulations with $\Delta t = 2$ could be used.

☐ simulations with just $\Delta t = 1$ are enough.

☐ simulations with $\Delta t = 1$ and one smaller one, for example $\Delta t = \frac{1}{2}$ are enough.

☒ simulations with $\Delta t = 1$ and at least both of the two smaller stepsizes are necessary. ✓



Answer

Correct:

The results for $\Delta t = 2$ are not valid, as has been demonstrated in the video.

With just $\Delta t = 1$ you do get the right answer, but because you have no idea of the error of the result, so you do not know whether the answer is right.

As there is no error estimation given for a horizontal error, just checking with one other stepsize might not be enough.

Because there is no error estimation given for a horizontal error, you have to see how the value converges numerically. You need at least three points to be able to see such a trend.

Submit

You have used 1 of 3 attempts

i Answers are displayed within the problem

Next, a page for questions you might have about this section.

© All Rights Reserved



© 2012-2017 edX Inc. All rights reserved except where noted. EdX, Open edX and the edX and Open EdX logos are registered trademarks or trademarks of edX Inc. | 粤ICP备17044299号-2

POWERED BY
OPENedX®

